A uniform harmonic limit for Hermite interpolation means

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Abstract

We prove Conjecture 22 of Horwitz's *Means and Hermite Interpolation*. For every fixed real power parameter, the means converge to the harmonic mean as the sum of the interpolation orders tends to infinity. The error is $O(1/n)$ uniformly over all admissible pairs of orders, including pairs whose difference stays fixed. The proof transforms a divided difference into a bounded moment of reciprocal nodes on a simplex, then couples the two moments whose ratio defines the mean.

Task 1: the source conjecture and its conventions

Fix $0 < a < b$ and nonnegative integers $m_2 < m_1$, and set

$$n = m_1 + m_2 + 1, \quad k = m_1 - m_2, \quad \alpha = m_1 + 1, \quad \beta = m_2 + 1, \quad d = n + 2 = \alpha + \beta + 1.$$

For a function f, let P, Q be the Hermite interpolants of degree at most n matching f through orders m_1, m_2 at a, b, respectively, with the orders interchanged for Q. Horwitz defines $M_{f,m_1,m_2}(a,b)$ as the unique $x \in (a,b)$ satisfying $P(x) = Q(x)$ when k is even, and $f(x) = (P(x) + Q(x))/2$ when k is odd.

For $p \notin \{0, 1, \dots, n\}$, write $M_{p,m_1,m_2} = M_{f,m_1,m_2}$ with $f(t) = t^p$. At $p = j \in \{0, 1, \dots, n\}$ the source explicitly defines the continuous extension using $f(t) = t^j \log t$. These conventions are stated immediately before Remark 5 of [1]; the conjecture is on its final page, p. 95.

Theorem 1. *For every fixed $p \in \mathbb{R}$ and every $a, b > 0$,*

$$\lim_{\substack{m_1+m_2\to\infty \\ 0\le m_2<m_1}} M_{p,m_1,m_2}(a,b) = H(a,b) = \frac{2ab}{a+b}.$$

More precisely, for $a < b$ let

$$q = -p - 1, \qquad c = \min_{1/b\le r\le 1/a} r^q > 0, \qquad L = \max_{1/b\le r\le 1/a} |q| r^{q-1}, \qquad C = \frac{L}{c}\left(\frac{1}{a} - \frac{1}{b}\right).$$

The value $L = 0$ is used when $q = 0$. Then for every admissible pair,

$$\left|\log \frac{(M_{p,m_1,m_2}(a,b) - a)b}{(b - M_{p,m_1,m_2}(a,b))a}\right| \le \frac{C}{n+2}, \tag{1}$$

and

$$|M_{p,m_1,m_2}(a,b) - H(a,b)| \le \frac{(b-a)C}{4(n+2)}. \tag{2}$$

The constants depend only on a, b, p, not on the split of the orders.

Symmetry handles $a > b$, and all means agree when $a = b$. The parameter p is fixed in this conjecture. A sequence of parameters growing with n, such as the source's arithmetic-mean specialization $p = n + 1$, is not covered by the asserted limit.

A reciprocal-node identity

Let $V = (V_1, \dots, V_d)$ be uniform on the simplex $\{v_j \ge 0 : \sum_j v_j = 1\}$. Equivalently, V has the Dirichlet distribution with all d parameters equal to one.

Lemma 1. *For positive nodes $z_1, \dots, z_d$ (repetitions allowed), set $N = d - 1$ and $B_N(p) = p(p-1)\cdots(p-N+1)/N!$. For $f_p(t) = t^p$,*

$$f_p[z_1, \dots, z_d] = B_N(p) \left(\prod_{j=1}^{d} z_j^{-1} \right) \mathbb{E} \left(\sum_{j=1}^{d} \frac{V_j}{z_j} \right)^{-p-1}. \tag{3}$$

At $p = j \in \{0, \dots, N-1\}$, the divided difference of $t^j \log t$ equals the right side with $B_N(p)$ replaced by $B_N'(j) \neq 0$ and the exponent replaced by $-j-1$.

Proof. The Hermite–Genocchi formula, obtained by iterating the fundamental theorem of calculus, is

$$f_p[z_1, \dots, z_d] = B_N(p) \mathbb{E} \left(\sum_{j=1}^{d} U_j z_j \right)^{p-N},$$

where U is uniform on the same simplex. The factor $1/N!$ in B_N accounts for its Euclidean coordinate volume $1/N!$. Repeated nodes follow by continuity.

Make the change of variables

$$R = \sum_{j=1}^{d} \frac{v_j}{z_j}, \qquad u_j = \frac{v_j/z_j}{R}.$$

It maps the simplex interior bijectively onto itself and gives $\sum_j u_j z_j = 1/R$. Its Jacobian in the first $d-1$ coordinates is

$$\left| \frac{\partial(u_1, \dots, u_{d-1})}{\partial(v_1, \dots, v_{d-1})} \right| = \frac{\prod_j z_j^{-1}}{R^d}. \tag{4}$$

For clarity, writing $r_j = 1/z_j$ and $R = r_d + \sum_{j<d}(r_j - r_d)v_j$, the derivative matrix is

$$\frac{1}{R} \operatorname{diag}(r_1, \dots, r_{d-1}) - \frac{1}{R^2} (r_j v_j)_{j<d} (r_j - r_d)_{j<d}^{\mathsf{T}}.$$

The determinant lemma gives the diagonal determinant times r_d/R, which proves (4). The transformed exponent is $R^{N-p-d} = R^{-p-1}$, proving (3).

At an excluded integer, differentiate the identity with respect to p. The simple zero of B_N makes its derivative the only surviving factor; the left side differentiates to the divided difference of $t^j \log t$. All differentiation is valid on a compact positive interval. This also verifies the source's extension without dividing by zero. □

The equation for the mean

Write $D_P = f[x, a^{[\alpha]}, b^{[\beta]}]$ and $D_Q = f[x, a^{[\beta]}, b^{[\alpha]}]$, where brackets denote node multiplicity, not numerical powers. The Hermite error formula gives

$$f(x) - P(x) = (x-a)^\alpha (x-b)^\beta D_P, \qquad f(x) - Q(x) = (x-a)^\beta (x-b)^\alpha D_Q.$$

In both parity cases the defining equation for the mean is

$$(x-a)^k D_P = (b-x)^k D_Q. \tag{5}$$

This is also equation (4) of [1].

For the power functions, including their integer extensions, define

$$A_P(x) = \mathbb{E}R_P(x)^q, \qquad A_Q(x) = \mathbb{E}R_Q(x)^q,$$

where R_P is the simplex average of the reciprocals of α copies of a, β copies of b, and one x; R_Q interchanges α and β. By Lemma 1, the divided differences have a common nonzero real factor, and hence

$$\frac{D_Q}{D_P} = \Big(\frac{a}{b}\Big)^k \frac{A_Q(x)}{A_P(x)}.$$

All quantities A_P, A_Q are strictly positive. Taking logarithms in (5) yields the exact identity

$$\log\frac{(x-a)b}{(b-x)a} = \frac{\log A_Q(x) - \log A_P(x)}{k}. \tag{6}$$

Coupling the moments, uniformly in the orders

Use the same uniform vector V for both averages, arranged as follows: β coordinates have coefficient $1/a$ in both averages, β coordinates have coefficient $1/b$ in both, and one coordinate has coefficient $1/x$ in both. On the remaining k coordinates, R_P uses $1/a$ and R_Q uses $1/b$. Permutation invariance of the uniform law gives exactly the required two marginal distributions. Consequently

$$R_P - R_Q = \left(\frac{1}{a} - \frac{1}{b}\right) T, \qquad T = \sum_{j \text{ in the } k \text{ differing coordinates}} V_j, \qquad \mathbb{E}T = \frac{k}{d}.$$

For every $x \in [a,b]$, both R_P and R_Q lie in $[1/b, 1/a]$. The function $g(r) = r^q$ is L-Lipschitz there and is at least c. Thus

$$|A_P - A_Q| \le L\mathbb{E}|R_P - R_Q| = L\left(\frac{1}{a} - \frac{1}{b}\right)\frac{k}{d}, \qquad \min(A_P, A_Q) \ge c,$$

and the mean value theorem for the logarithm gives

$$\frac{|\log A_Q - \log A_P|}{k} \le \frac{L}{c}\left(\frac{1}{a} - \frac{1}{b}\right)\frac{1}{d} = \frac{C}{n+2}. \tag{7}$$

Together with (6), this proves (1). The cancellation of k is why the estimate holds even when $k = 1$.

To pass to an ordinary distance, put $r = (a/b)e^y$ and

$$\Phi(y) = \frac{a + br}{1 + r}.$$

Then $\Phi(0) = H(a,b)$ and $0 < \Phi'(y) = (b-a)r/(1+r)^2 \le (b-a)/4$ for all real y. Apply this global Lipschitz bound to $y = \log((x-a)b/((b-x)a))$. It gives (2) and completes the proof of Theorem 1.

Corollary 1. *For $a < b$, $M_{p,m_1,m_2}(a,b)$ is strictly above $H(a,b)$ if $p > -1$, strictly below it if $p < -1$, and equal to it if $p = -1$.*

Proof. In the coupling $R_P > R_Q$ almost surely. Thus $A_P > A_Q$ for $q > 0$, and $A_P < A_Q$ for $q < 0$. Use (6) and the strict increasingness of Φ. If $p = -1$, then $q = 0$ and both moments are one. The last case recovers the source's Theorem 7. □

Follow-up tasks

These directions follow the full solution of Task 1; they are not additional completed or independently screened open problems.

Task 2: Compute the first nonzero asymptotic term of $M_{p,m_1,m_2} - H$ when $k/n \to \lambda \in [0,1]$. Treat the regime $\lambda = 0$ uniformly, including both bounded and unbounded differences k.

Task 3: Classify the limits when the parameter also varies, especially $p_n/n \to \theta$. Relate the resulting limits to the source's geometric and arithmetic specializations, and identify transitions between regimes.

Task 4: Give verifiable conditions on an analytic function f under which its Hermite means converge to an arithmetic or a shifted harmonic mean. Determine the role of the position and order of its nearest singularities.

Task 5: Construct an algorithm with certified errors for high-order Hermite means using the bounded reciprocal moments. Obtain explicit complexity bounds uniform over the order split and compact sets of p.

Status and checks

The accompanying dated audit records live MathDB status and searches for earlier solution claims. The current proof is internally checked, not externally peer reviewed. Its error bound is analytic; the accompanying exact and numerical tests only check formulas and examples.

References

[1] A. Horwitz, *Means and Hermite Interpolation*, Journal of Mathematical Inequalities **2**(1) (2008), 75–95, Conjecture 22. `https://doi.org/10.7153/jmi-02-08`. Preprint: arXiv:0711.4940v2, `https://arxiv.org/abs/0711.4940`.